

IOT system development for Dissolved Oxygen measurement



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2023CHB1082

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Problem Statement

- Conventional dissolved oxygen measurement methods are **manual, time-consuming**, and provide only **discrete measurements**.
- Although sensor-based DO meters enable continuous monitoring, most commercially available instruments are expensive and often inaccessible for many sectors that require frequent water quality monitoring.
- In addition, these systems typically provide localized measurements without integrated remote data access or long-term monitoring capabilities.
- Therefore, there is a need to develop a **cost-effective dissolved oxygen sensing system** integrated with **IoT-based monitoring**, enabling continuous measurement and wider accessibility.

Dissolved Oxygen

Dissolved Oxygen (DO) is the amount of molecular oxygen dissolved in water which is expressed in mg/L (ppm) or % saturation. It is essential for survival of aquatic organisms, aerobic biological reactions and in wastewater treatment.

The factors affecting DO are:

- Temperature
- Salinity
- Pressure
- Mixing and flow conditions

Working Principle of DO Sensor

Materials Used:

- Cathode: Gold (Au) electrode
- Anode: Silver (Ag) electrode
- Electrolyte: Potassium Hydroxide (KOH)
- Membrane: Silicone oxygen permeable membrane

At cathode: $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$

At anode: $4Ag + 4OH^- \rightarrow 2Ag_2O + 2H_2O + 4e^-$

The current is converted to DO using the following equation

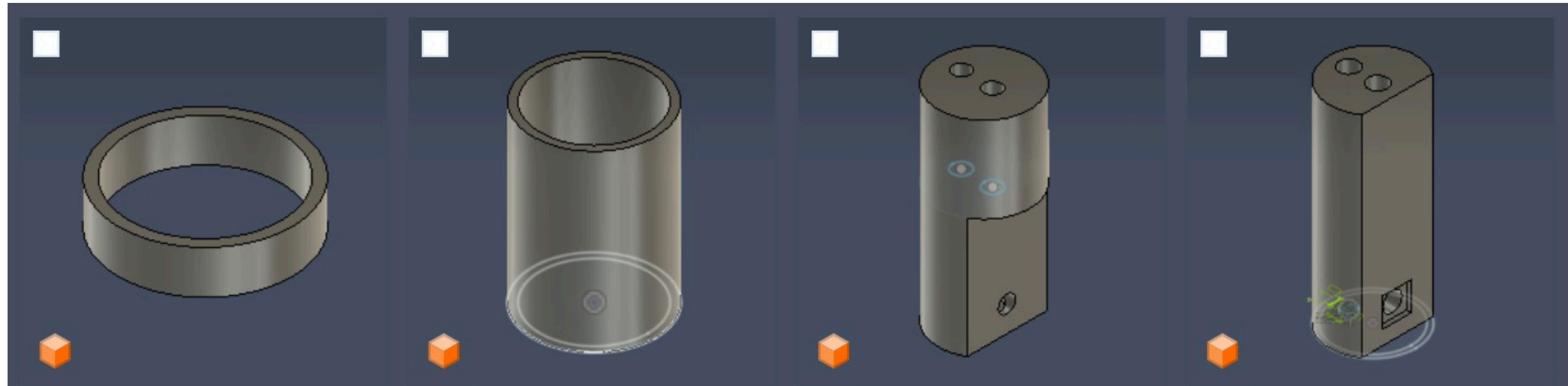
$$C_{O_2} = \frac{I - I_o}{S}$$

Where:

S → sensor sensitivity

I_o → zero-oxygen offset current

Sensor Fabrication



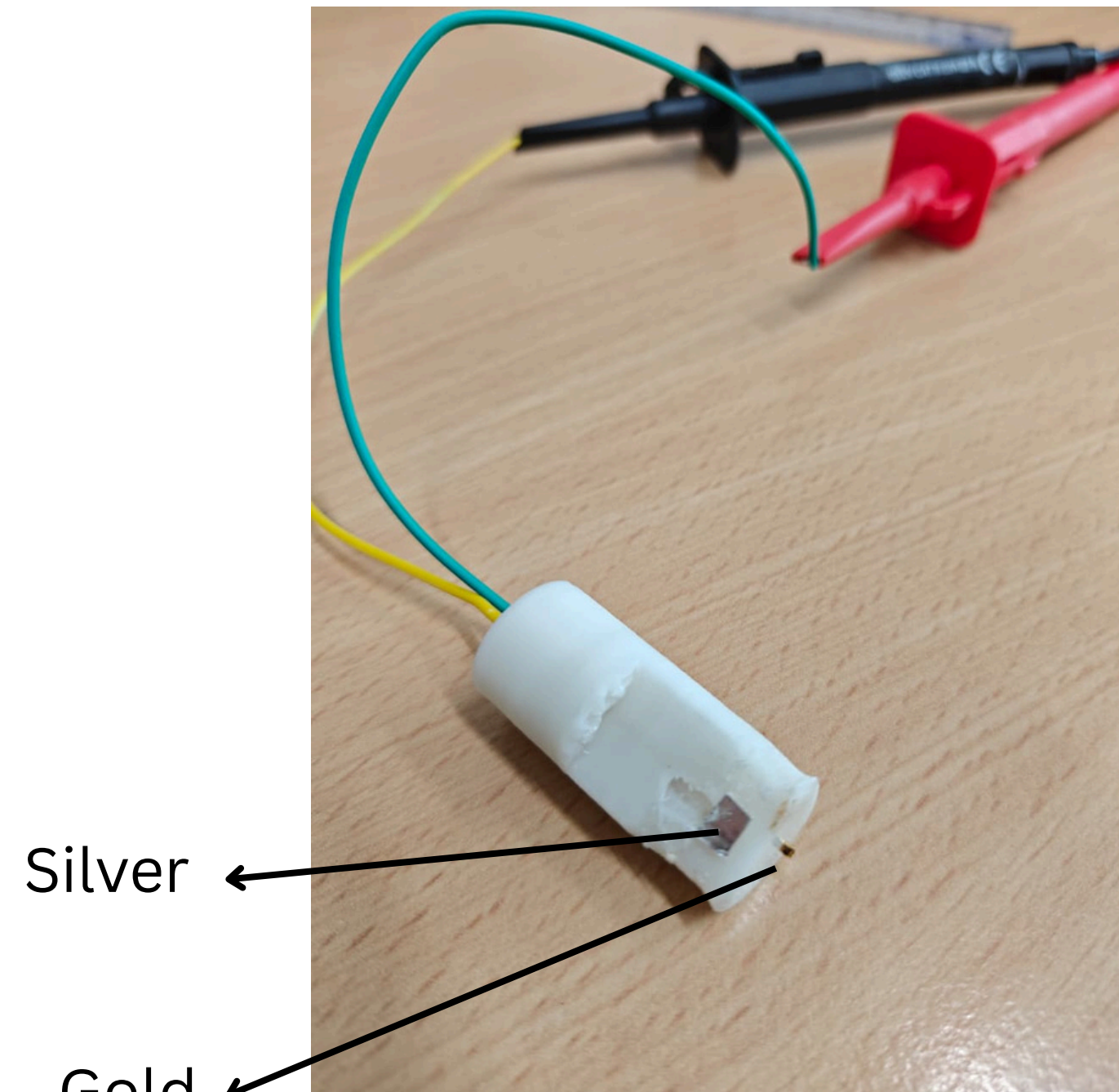
The sensor consists of 4 main components:

- Membrane
- Outer Casing
- Electrode Support
- Retaining Ring

Sensor Fabrication



Assembled sensor



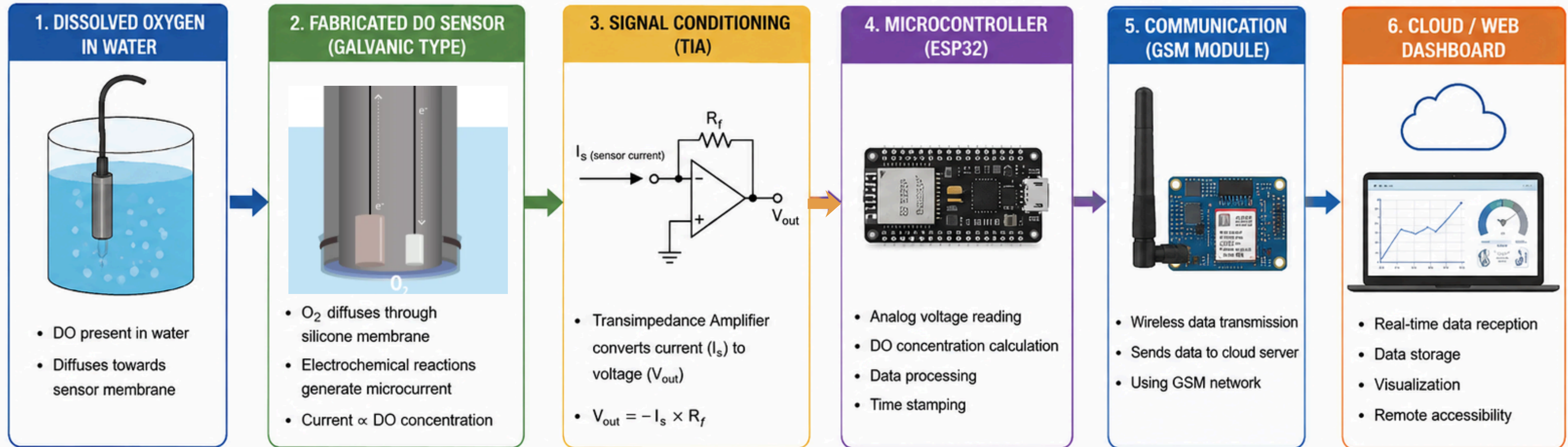
Silver

Gold

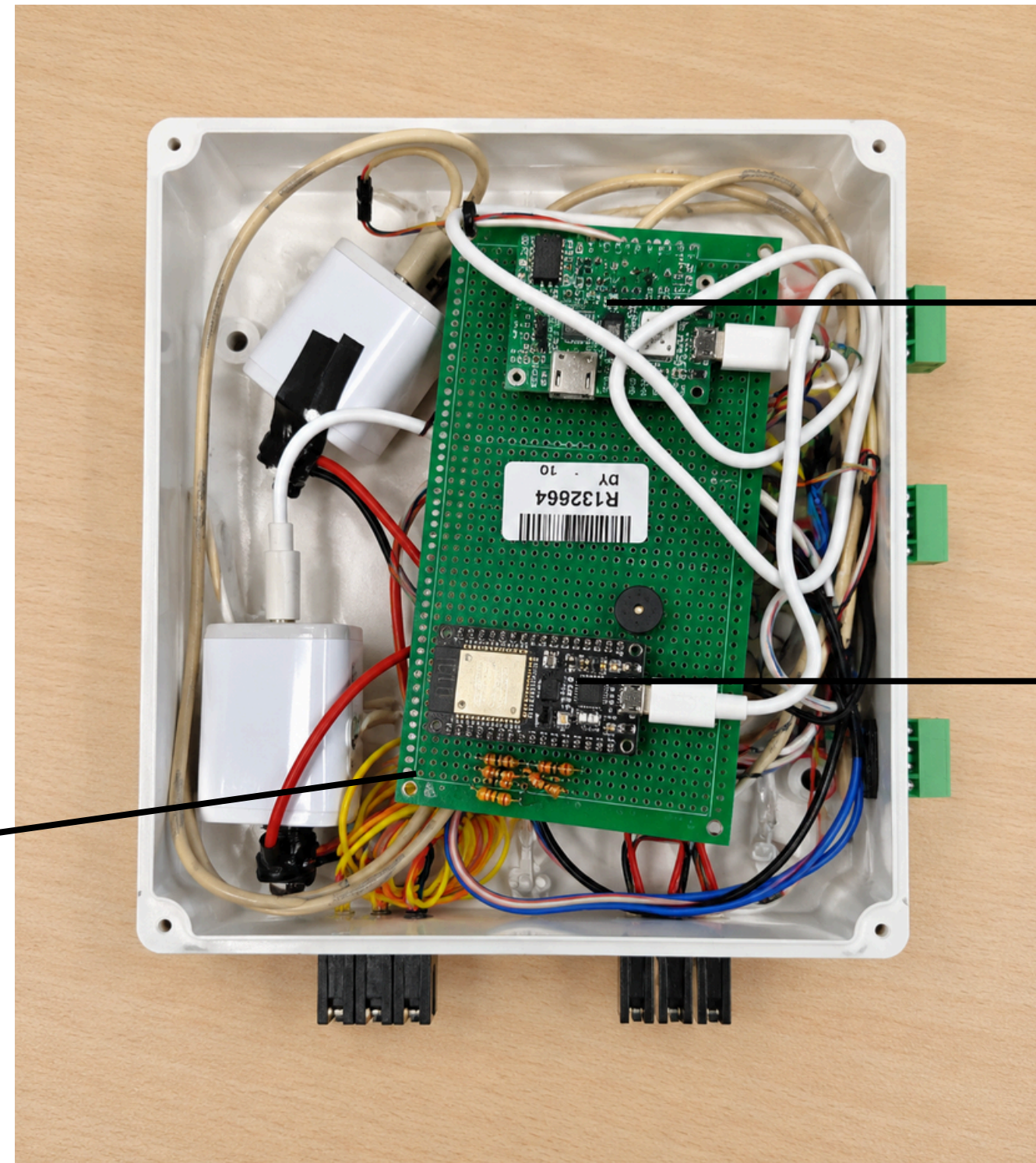
Sensor with connections

System Architecture

IoT Based Dissolved Oxygen Monitoring System



IoT System Setup



Key Functions:

- Acquires voltage signal from DO sensor
- Converts and processes signal using ESP32
- Performs DO calculation using calibration
- Transmits data to cloud via GSM module
- Enables real-time monitoring and analysis

Vitus Nanobubble Generator

Why Nanobubble Integration?

- Enhances dissolved oxygen concentration
- Produces fine bubbles with high surface area
- Improves gas-liquid mass transfer
- Helps study oxygen transfer behaviour dynamically

Normal Aeration

Large bubbles
Low surface area
Fast escape
Low OTE
Moderate kLa

Nanobubble System

Nano-sized bubbles
Very high surface area
Remain suspended
High OTE
High kLa

Mass Transfer Analysis

Volumetric Mass Transfer coefficient($k_L a$)

$$\frac{dC}{dt} = k_L a (C^* - C)$$
$$\ln \left(\frac{C^* - C_0}{C^* - C_t} \right) = k_L a t$$

Indicates

- Oxygen transfer rate
- Aeration performance
- Dynamic oxygen transfer behaviour

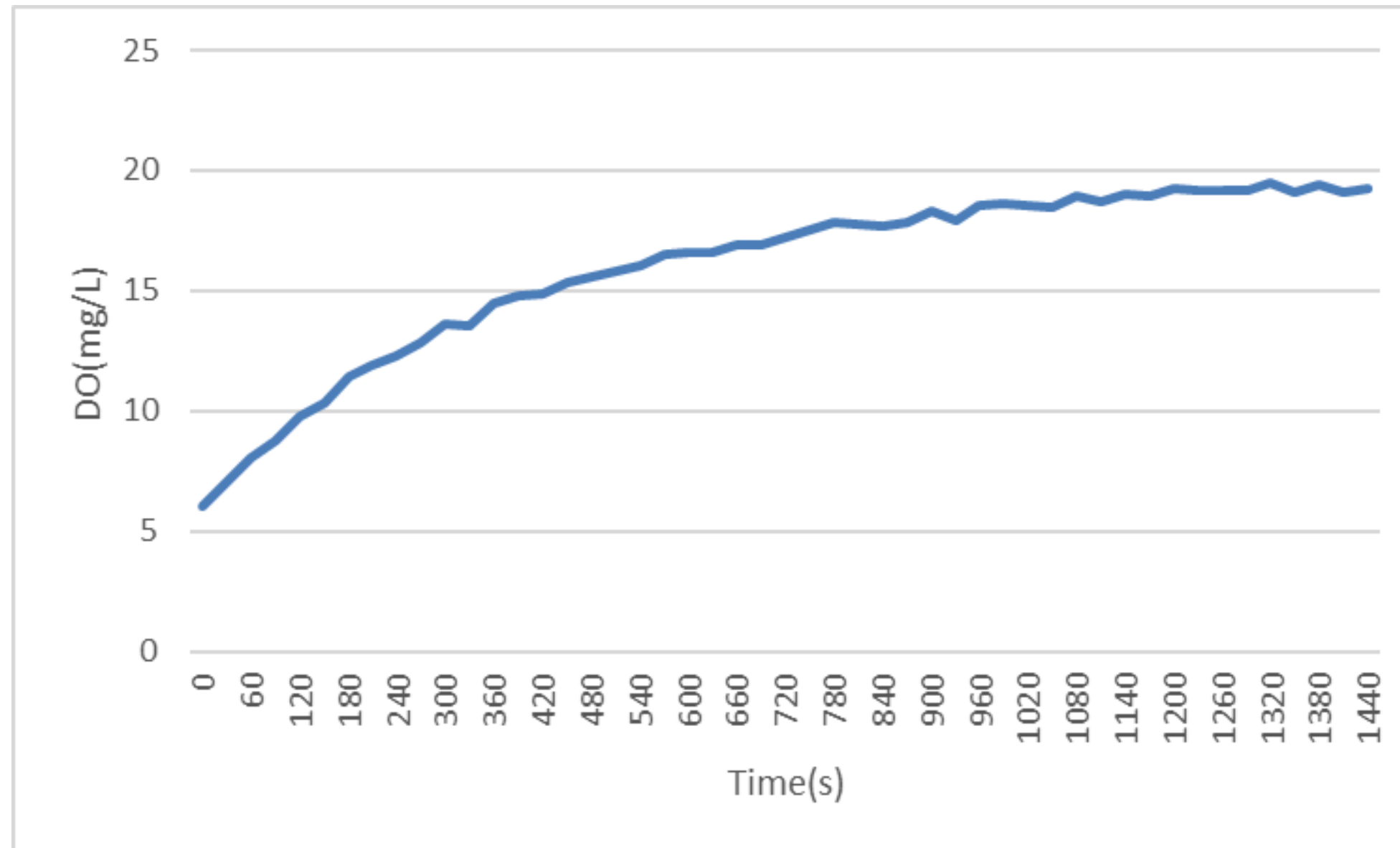
Oxygen Transfer Efficiency(OTE)

$$OTE = \frac{(C_t - C_0)V}{Q \times t \times C_{gas} \times 1429} \times 100$$

Indicates

- Oxygen utilization efficiency
- Overall aeration effectiveness
- Transfer efficiency of supplied gas

DO Trends

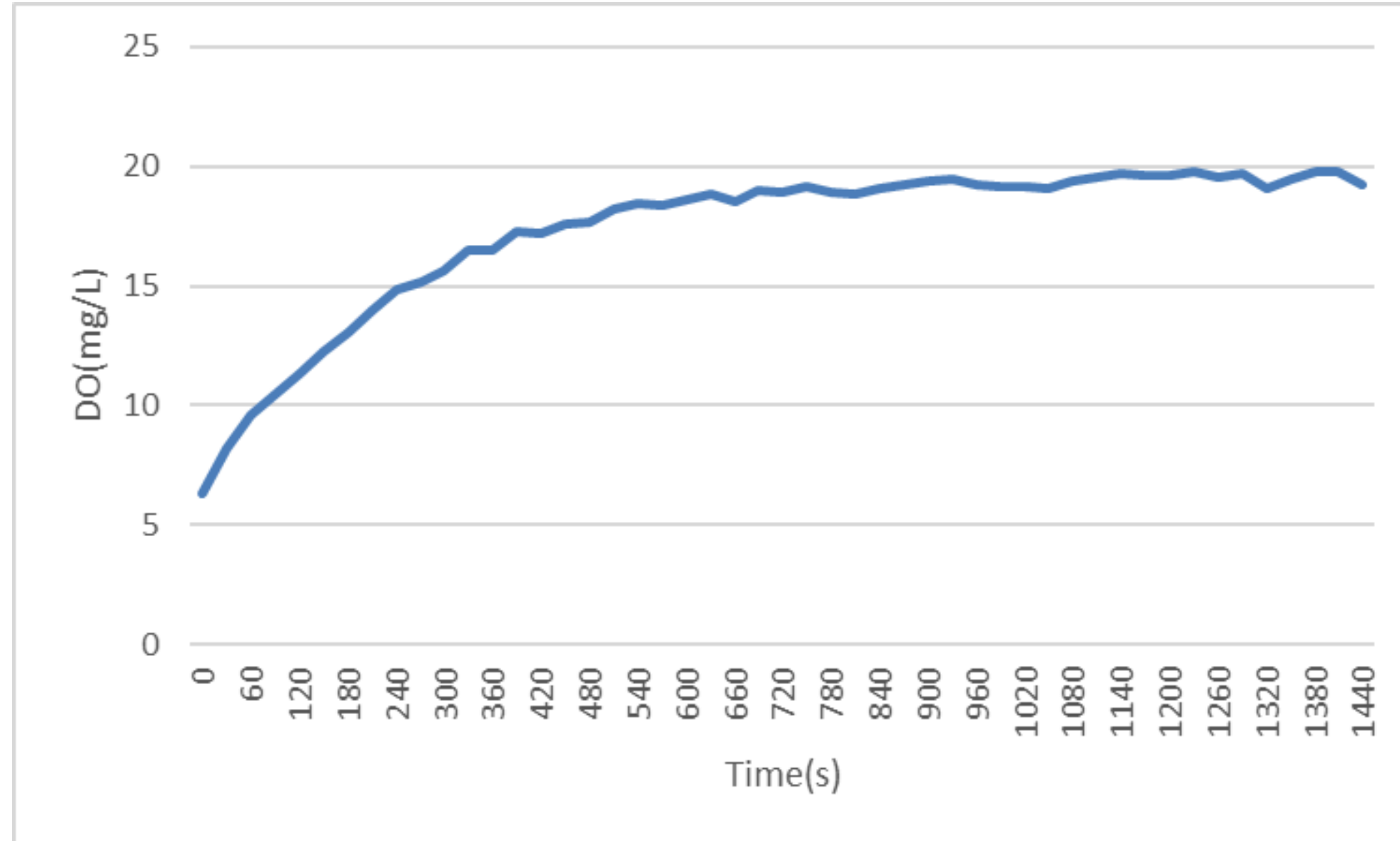


$$k_L a = 0.0021 \text{ min}^{-1}$$

$$\text{OTE} = 12.45\%$$

Case 1: Slow Oxygen Transfer Behaviour

DO Trends

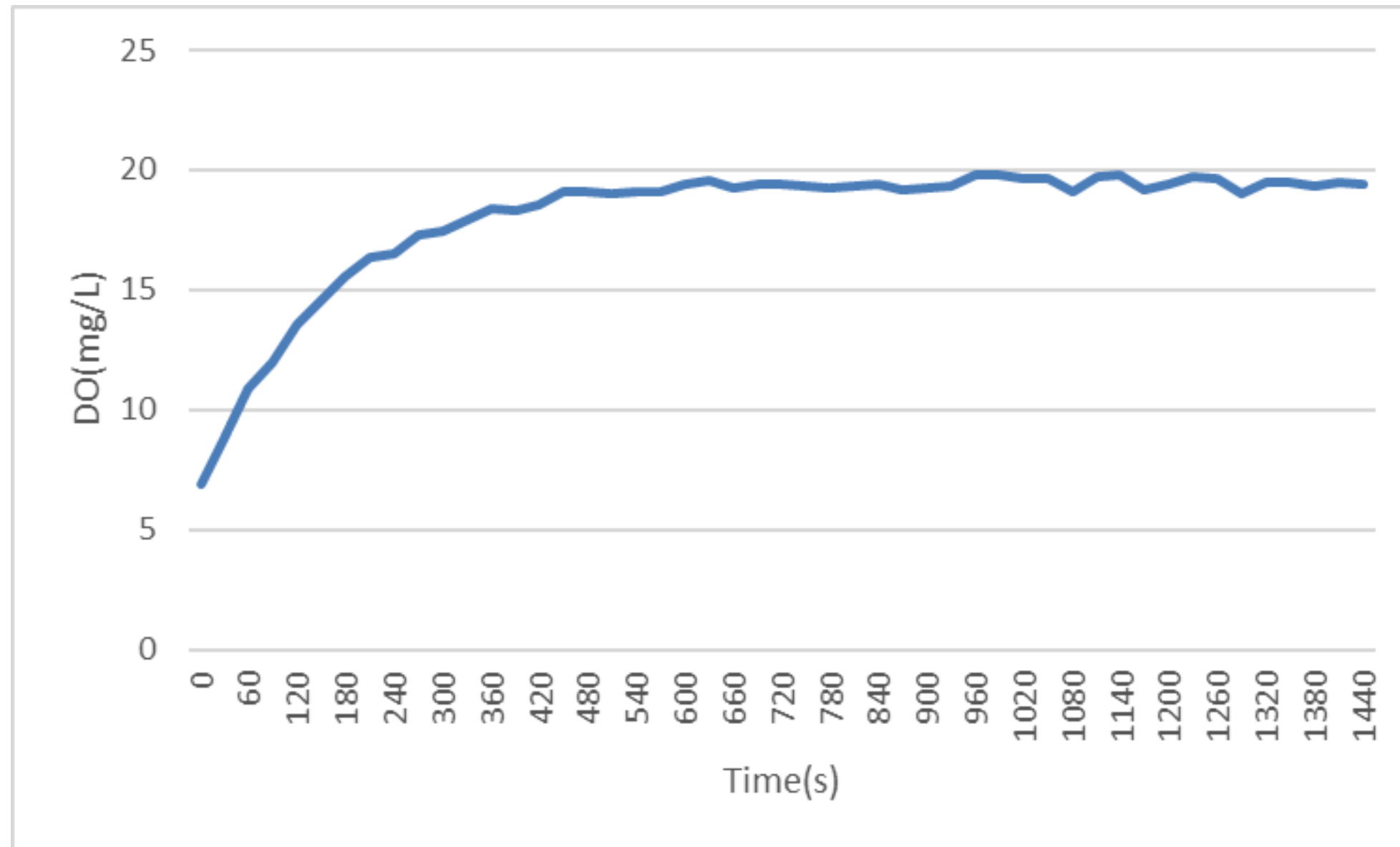


$$k_L a = 0.0038 \text{ min}^{-1}$$

$$\text{OTE} = 12.23\%$$

Case 2: Moderate Oxygen Transfer Behaviour

DO Trends

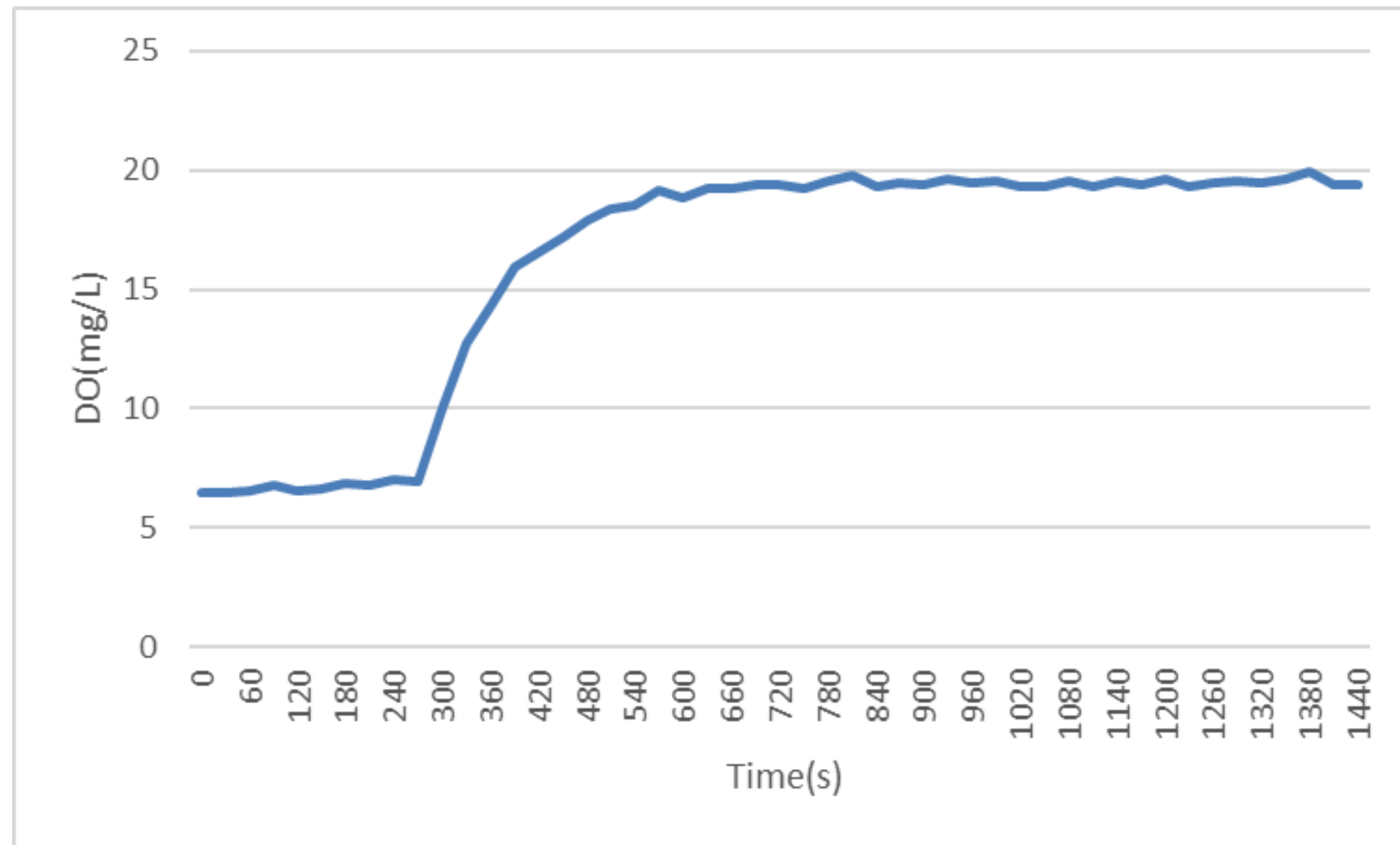


$$k_L a = 0.0053 \text{ min}^{-1}$$

$$\text{OTE} = 11.38\%$$

Case 3: Enhanced Oxygen Transfer Behaviour

DO Trends

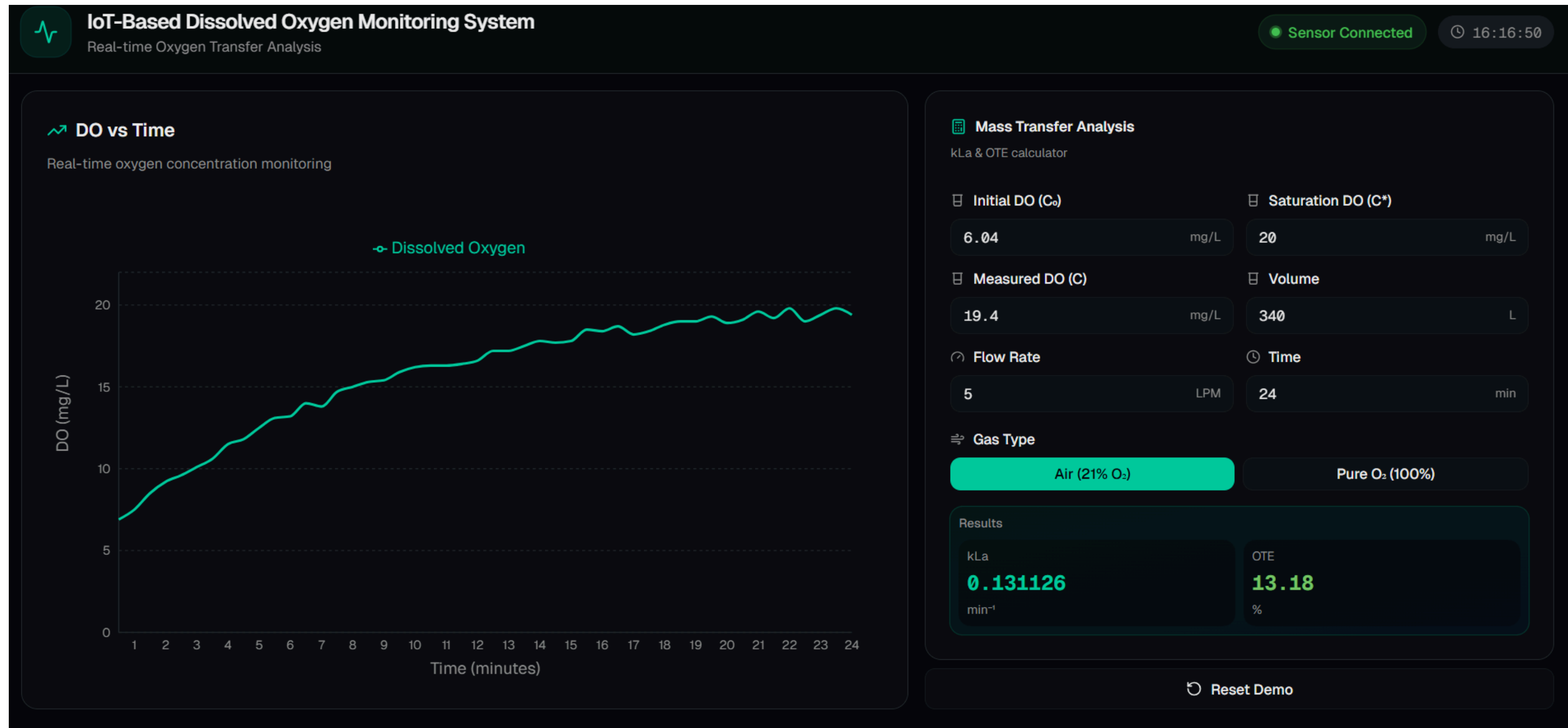


$$k_L a = 0.007 \text{ min}^{-1}$$

$$\text{OTE} = 12.15\%$$

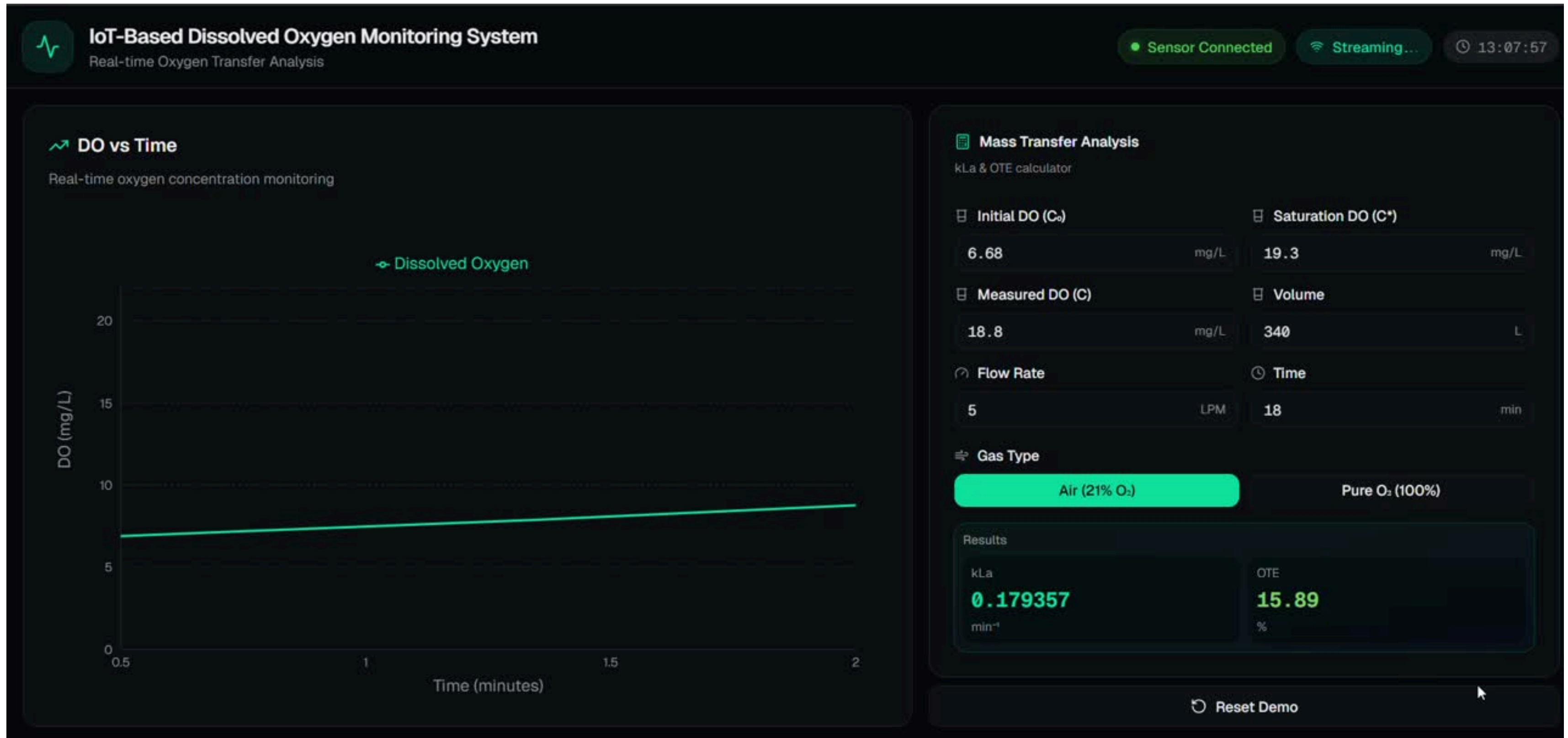
Case 4: Delayed Aeration Response

Web interface



Link : <https://dissolvedoxygenmonitor.vercel.app/>

Demonstration



Future Scope

- The measurement range of the developed sensor can be further improved to accurately detect dissolved oxygen levels beyond 20 mg/L.
- The stability and accuracy of the sensor can be enhanced.
- The system can be extended to include automated control strategies, where aeration can be adjusted based on real-time dissolved oxygen levels.
- The developed setup can be scaled and adapted for applications in industrial processes and wastewater treatment systems.

References

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Online Intelligent detection
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THANK YOU